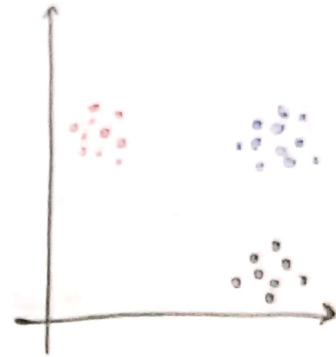


* Embedding

Turning Meanings into Numbers.

What are Embeddings?

convert anything (text, image, audio, behaviour) into vectors so machines can understand relationship.



King: $[-0.25, -0.13, 0.78, \dots]$

Queen: $[0.27, -0.10, 0.80, \dots]$

Apple: $[-0.60, 0.90, -0.12, \dots]$

similar things are close.
Different things are far.

$\Rightarrow$ King $\approx$ Queen. (close vectors)

$\Rightarrow$ King $\neq$ Apple. (far vectors)

How Embedding are Computed?

Embedding are computed by feeding data into

- ① Neural Network (Word2Vec, BERT)
- ② Matrix factorization (e.g. ~~SVD~~ SVD, PCA)
- ③ Triplet Loss Learning.
- ④ Recommendation Models.

Steps to compute

- ① Preprocessing Data
- ② Initialization
- ③ Training.
- ④ Extracting the Vector

$$f(x) \rightarrow R^d$$

x = input (word, sentence, image)

d = embedding dimension

(e.g. 128, 768, 1536)

Why Embedding?

↳ Traditionally we use One Hot Encoding.

✗ Huge dimensionality

✗ No semantic meaning.

↳ With embedding we capture :

✓ Semantic similarity.

✓ Context

✓ Relationships

✓ Patterns in Data.

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Similarity Measures.

- ① Cosine Similarity - (Most Common)

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

1 → identical.

0 → unrelated

-1 → opposite

- ② Euclidean Distance

straight line distance in space



Types of Embedding. — **

- ① Word Embedding → Word2vec, GloVe, FastText
paris $\approx$ France or Delhi $\approx$ India
- ② Sentence Embedding →
 - I love AI
 - I enjoy Artificial intelligence } → similar vectors
- ③ Image Embedding → Image to vectors using CNN
- ④ Multimodal Embedding → same vector space for
text + image + audio.

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Real world Use-Case. — **

- ↳ Semantic Search.
- ↳ Recommendation System
- ↳ RAG (Retrieval Augmented Gen)
- ↳ Clustering & Classification.
- ↳ Anomaly Detection